

AMTI (NMTC) - 2014

RAMANUJAN CONTEST - FINAL - INTER LEVEL

1. ABC is a triangle in which $AB > AC > BC$. D is a point on the minor arc BC of the circumcircle of the triangle ABC. O is the circumcentre. E and F are the intersection points of the line AD with the perpendiculars from O to AB and AC respectively. P is the point of intersection of BE and CF. If $PB = PC + PO$, find the angle A of the triangle ABC.
2. a) For the positive integer n define $f(n) = 1^n + 2^{n-1} + 3^{n-2} + \dots + (n-2)^3 + (n-1)^2 + n^1$.
What is the minimum value of $\frac{f(n+1)}{f(n)}$?
b) ABC is an isosceles triangle in which $AB = AC$. The bisector of $\angle B$ meets AC at D and it is given that $BC = BD + AD$. Find $\angle A$ of the triangle ABC.
3. Let n be a positive integer and S_n be the set of all positive integer divisors of n (including 1 and itself). Prove that at most half of the elements of S_n have their units digit equal to 3.
4. a) Let A be a set of 8 elements. Find the maximum number of 3 – element subsets of A, such that the intersection of any two of them is not a 2 element set.
b) a, b, c, d, are all positive reals and $\frac{1}{1+a^4} + \frac{1}{1+b^4} + \frac{1}{1+c^4} + \frac{1}{1+d^4} = 1$. Prove that $abcd \geq 3$.
5. In a plane there are two similar, convex quadrilaterals ABCD and $AB_1C_1D_1$ such that C, D are inside $AB_1C_1D_1$ and B is outside $AB_1C_1D_1$. Prove that if the lines BB_1, CC_1, DD_1 are concurrent, then ABCD is cyclic. Is the converse true ?
6. Prove that if the integer n is not divisible by 5, then the polynomial $f(x) = x^5 - x + n$ cannot be factored as the product of two non-constant polynomials with integer coefficients.
7. a) One may perform the following two operations on a positive integer.
(i) Multiply it by any positive integer.
(ii) Delete zeros in its decimal representation.
b) Show that $1! + 2! + 3! + \dots + 2013!$ can not be written as n^k for any integer n and integer $k \geq 2$.
8. a) $\lfloor x \rfloor$ denotes the floor function (the greatest integer function). Let r be a real number for which $\left\lfloor r + \frac{19}{100} \right\rfloor + \left\lfloor r + \frac{20}{100} \right\rfloor + \left\lfloor r + \frac{21}{100} \right\rfloor + \dots + \left\lfloor r + \frac{91}{100} \right\rfloor = 546$. Solve the equation $x + \lfloor 100r \rfloor = 2013$.
b) For all distinct positive integers m and n prove $(2013)^{2^n} + 2^{2^n}$ is relatively prime to $(2013)^{2^m} + 2^{2^m}$.
